A Pneu Robot

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Project Idea and Intellectual Merit

The project was inspired by IIT - Italian Institute of Technology

Bio-inspiration: Earthworms

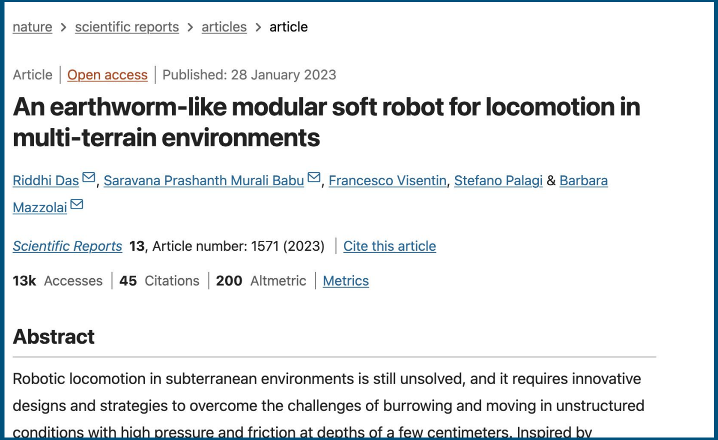
“Space Crawler”: a soft robot which when actuated (pressurised) crawls across the floor. Developing the design to include a gripper that folds in using a PneuNet actuator link.

Our idea exploited the following features of soft robotics:

- Gripper to grab items.
- Body to crawl across the floor.

One challenge we had was fabrication

For the future consider REGOLITH and closed-loop fluid actuation!



nature > scientific reports > articles > article

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An earthworm-like modular soft robot for locomotion in multi-terrain environments

Riddhi Das, Saravana Prashanth Murali Babu, Francesco Visentin, Stefano Palagi & Barbara Mazzolai

Scientific Reports **13**, Article number: 1571 (2023) | Cite this article

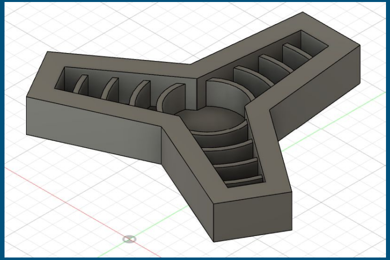
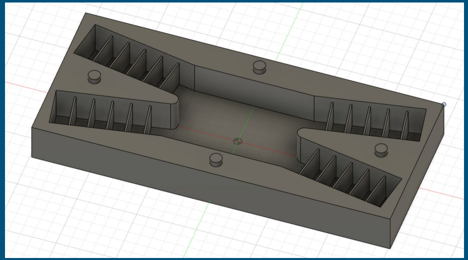
13k Accesses | **45** Citations | **200** Altmetric | Metrics

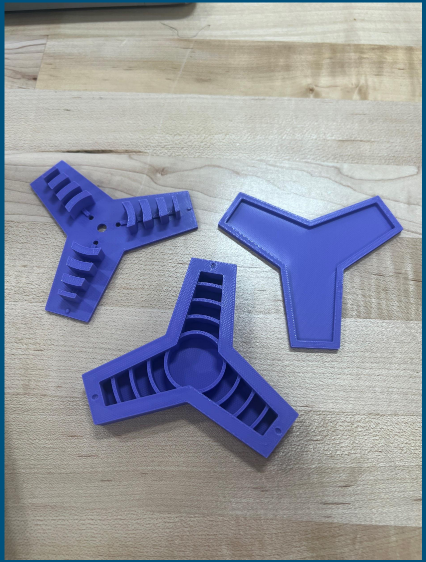
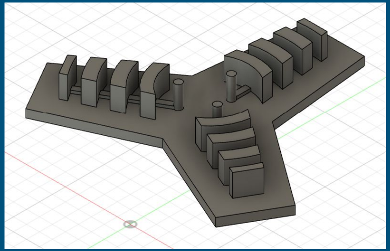
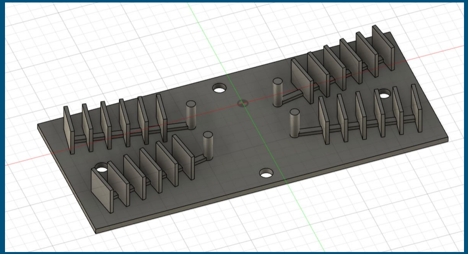
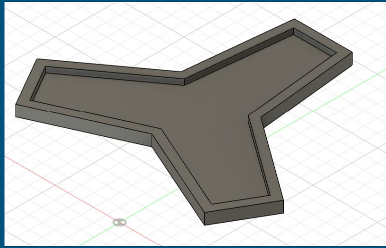
Abstract

Robotic locomotion in subterranean environments is still unsolved, and it requires innovative designs and strategies to overcome the challenges of burrowing and moving in unstructured conditions with high pressure and friction at depths of a few centimeters. Inspired by

Design and Manufacturing - Molds

3D printed - PLA molds

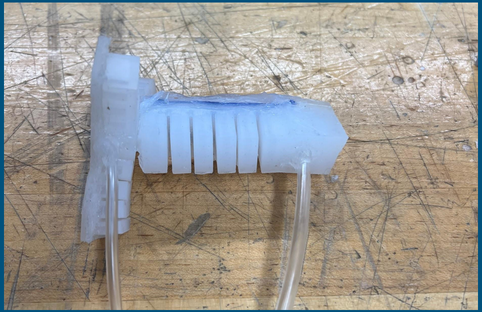





Design and Manufacturing - Materials

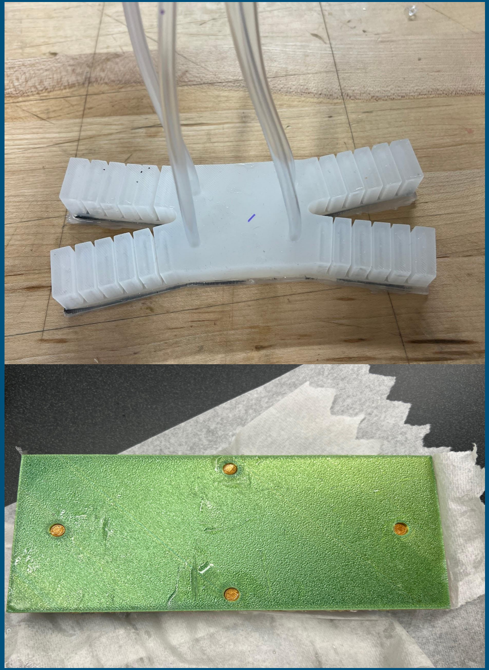
Dragon Skin 10

Plastic Tubing

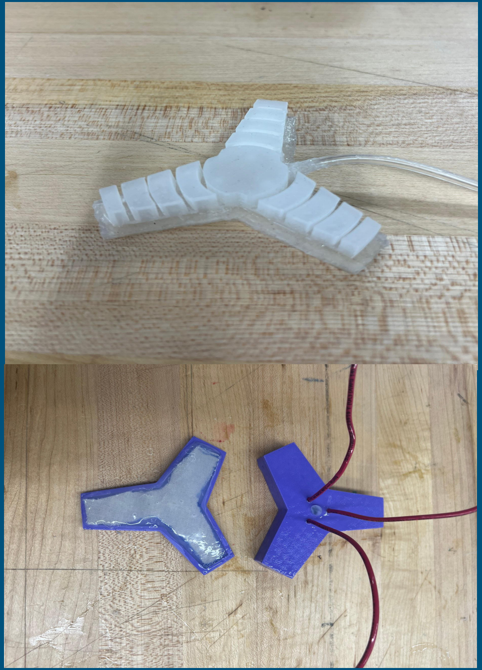
Connective Piece



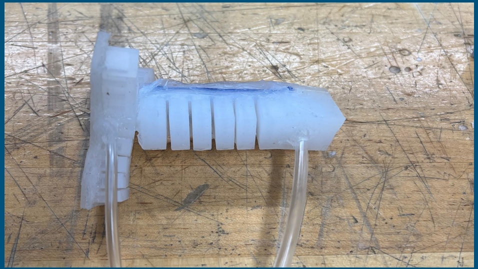
Base



Gripper



Design and Manufacturing - Integration



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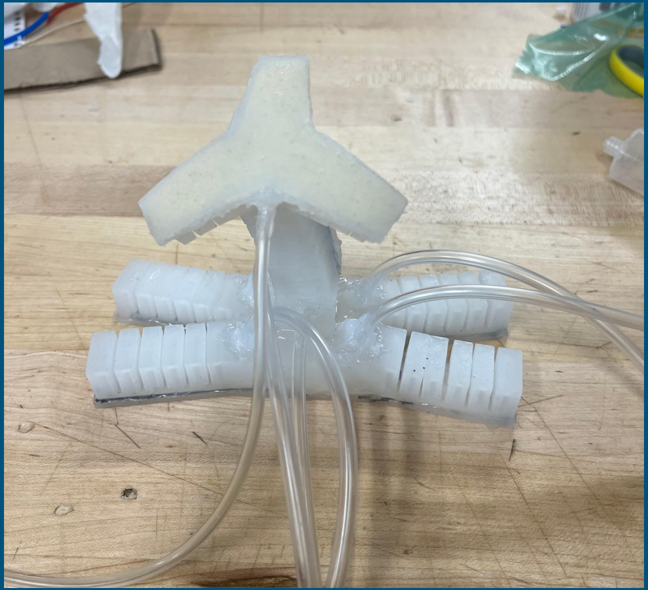


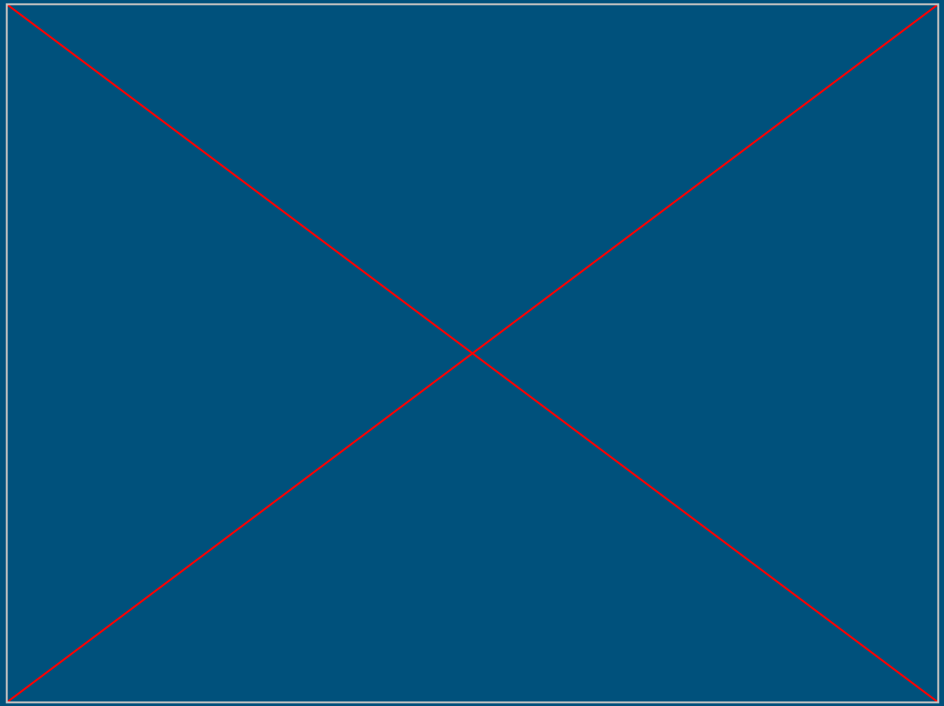
Testing/ Validation of Functionality

Speed: 1 body length (17cm) / 90 seconds

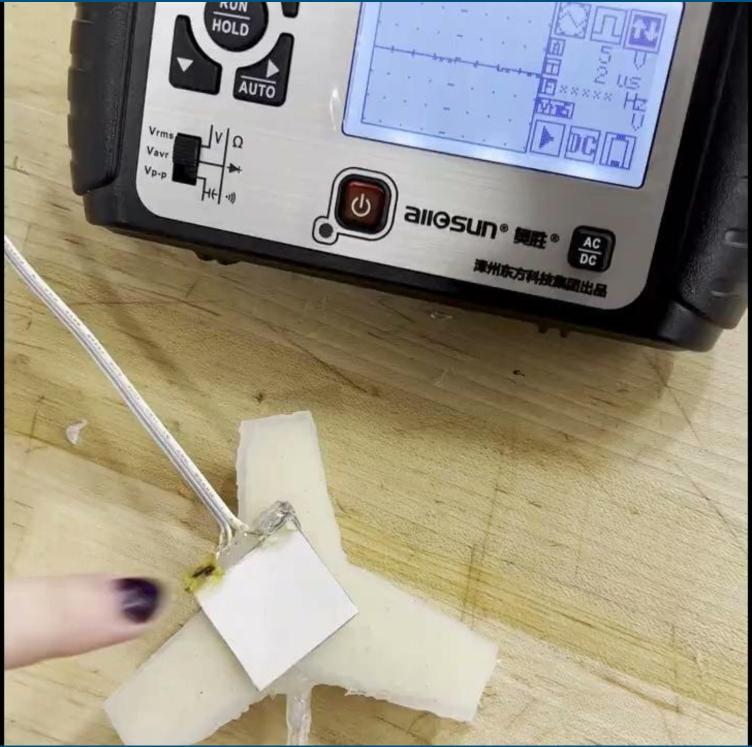
Max slope body can climb: ~ 15 degrees

Max diameter to squeeze to: 6cm

Testing - Walking



Testing - Contact Sensor

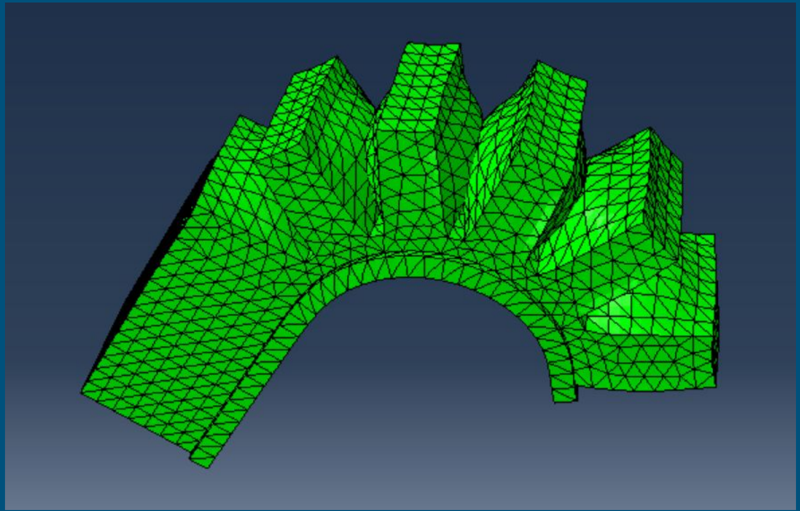


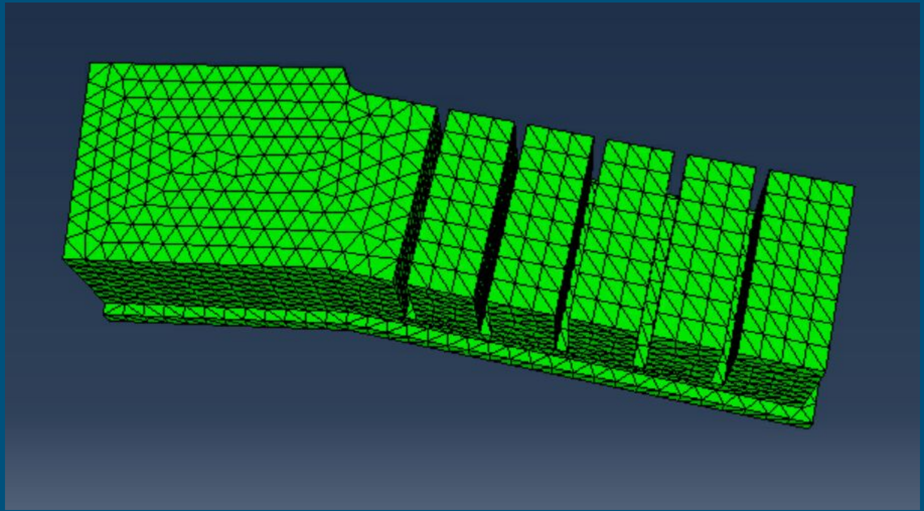
Testing - Gripper

Testing - Gripper With Sensor

Testing - Gripper Arm

Modeling Component





Conclusion and Broader Impacts

Successful fabrication of a soft robot

Design and assembly trial and error

Understanding of pneu-net actuators

Useful for uneven terrain and inclement weather

Doesn't have exposed hardware components like rigid robots, could be used to explore other worldly terrains

Exploration in hazardous environments, durable for unknown areas, retrieve materials from disaster zones

Environmental sampling, monitor ecosystems in sensitive habitats

Add Satae (like earthworms) to move and grio

Contributions

Parker - Main body CAD, molds, fabrication , and modeling

Christina - Gripper CAD, molds, and fabrication

Anthony - Sensor design and robot integration

Anisa - Robot application, robot integration, and robot characterization

Elliot - Arm design, robot integration, and robot characterization

References

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https://www.universetoday.com/160388/an-earthworm-robot-could-help-us-explore-other-worlds/